CHERI FAVEL



Chemical Engineer (MEng) | Process Engineering & Industrial R&D

Abu Dhabi, UAE | UAE resident | Available immediately

cherifavel@gmail.com | +971 58 584 5473 | linkedin.com/in/cheri-favel | Engineering portfolio

PROFESSIONAL PROFILE

MEng Chemical Engineer with industrial and postgraduate research experience in process development, hydrometallurgy and electrochemical systems. Hands-on experience in electrowinning, selective precipitation, metal recovery and industrial water treatment, supported by mass balances, process documentation and technical reporting. Published researcher seeking process engineering, applications or industrial R&D roles in the UAE, particularly where electrochemistry, metal recovery and water chemistry apply, such as energy transition, critical minerals, catalyst recovery and industrial water.

TECHNICAL CAPABILITIES

- **Process engineering:** mass and energy balances, process calculations, PFDs/P&IDs, equipment-sizing support, design bases, specifications and supplier quotation review.
- **Process development:** experimental design, leaching, selective precipitation, electrowinning, laboratory reactor testing and technical-economic evaluation support.
- **Laboratory and data:** ICP-OES, XRF, XRD, ATR-FTIR, SEM-EDS, BET, TGA, pH and particle-size analysis; kinetic modelling, data interpretation and technical reporting.
- **Software:** Excel, VBA, Power Query, PivotTables, Visio and Python; foundational Aspen Plus/HYSYS exposure.

ENGINEERING AND RESEARCH EXPERIENCE

Process & R&D Engineer Mar 2021 – Jul 2022

Free Radical Process Design (FRPD), South Africa

- Conducted laboratory and engineering investigations for hydrometallurgical, electrochemical and industrial water-treatment process development.
- Prepared process calculations, mass and energy balances, flow diagrams (PFDs/P&IDs) and technical reports; reviewed and revised design bases; supported equipment sizing and supplier quotation review.
- **Manganese processing:** contributed to design-basis and process documentation for a manganese sulphate monohydrate demonstration plant; ran selective-precipitation investigations.
- **Electrowinning:** ran copper and cobalt trials on a continuous rotating-cathode unit, investigating deposit formation and release; contributed to a proposed zinc-recovery application.
- **Zinc electrolysis:** planned, set up, ran and reported plate-electrode zinc electrolysis trials end to end, including experimental design, sampling, analysis and recommendations.
- **Gold recovery:** ran laboratory electrowinning/sludging reactor trials on refinery eluate, varying current density, voltage, flow rate and residence time; supported mass-balance and ICP-OES/XRF data analysis.
- **Rare earths:** ran lanthanide precipitation experiments and contributed calculations to a CAPEX/OPEX model for a neodymium metal plant.
- **Industrial water:** investigated metal removal from synthetic mine-tailings water using oxidation and selective precipitation; wrote the laboratory-trial section of a research proposal for Fe, Zn and Mn recovery from Peruvian tailings water.

Postgraduate Researcher in Chemical Engineering Jan 2020 – Dec 2021

University of Pretoria, South Africa

- Designed nitric-acid leaching experiments on phlogopite and characterised mineral feed and residues; identified diffusion-controlled kinetics (one-dimensional D1 model) and determined activation energies; research published in *Clays and Clay Minerals* (2022).
- Planned laboratory work, maintained experimental records, guided undergraduate laboratory work and communicated results through scientific reports.

ADDITIONAL PROFESSIONAL EXPERIENCE

Business Setup & Marketing Consultant Apr 2025 – Present
Cresco Consulting FZCO, UAE

- Prepare and submit company-registration documentation through the IFZA portal; manage client information and records; maintain the company website and creative content.

Engineer Assistant / Deckhand / Stewardess May 2023 – Apr 2025
Research Vessel Rosa, United States & Bahamas

- Assisted with pumps, plumbing and RO watermaker maintenance and troubleshooting; organised technical drawings, maintenance records and procurement within marine safety procedures.

Academic Journal Editor Nov 2022 – Apr 2023
CACTUS Communications, remote

- Edited scientific manuscripts for technical clarity, consistency and journal requirements; reviewed terminology and provided structured author feedback.

Teaching Assistant (Part-Time) Feb 2019 – Jun 2021
University of Pretoria, South Africa

- Supported laboratory sessions and tutoring in particle technology, kinetics and engineering practice/costing; supervised practical work and assessed student submissions.

Online English Language Consultant Sep 2020 – Mar 2021
MPC Connect, South Africa / remote

- Delivered structured lessons and feedback to non-native English speakers, developing clear communication and independent remote-working skills.

Laboratory Intern (Vacation Work) Dec 2017 – Jan 2019
University of Pretoria, South Africa

- Completed 12 weeks of laboratory placements in mineral leaching and fertiliser production, including experimental optimisation and analytical testing.

EDUCATION AND RESEARCH

MEng Chemical Engineering | University of Pretoria | 2021

BEngHons Chemical Engineering with distinction | University of Pretoria | 2020

BEng Chemical Engineering | University of Pretoria | 2019

- **MEng dissertation:** Development of Kinetic Parameters for Nitric Acid Leaching of Phlogopite and Characterization of the Solid Residue.
- **Publication:** Favel, C.M. & du Plessis, B.J. (2022). Development of Kinetic Parameters for Nitric Acid Leaching of Phlogopite and the Characterization of Solid Products. *Clays and Clay Minerals*, 70, 106–119. doi.org/10.1007/s42860-022-00180-x
- **Environmental Engineering specialisation:** water and effluent treatment, bioreactors, air pollution, waste remediation and environmental regulation.

ResearchGate: researchgate.net/profile/Cheri-Favel

RELEVANT PROFESSIONAL DEVELOPMENT

- **SAMA Return-to-Work Programme, Phase 2** | Selected participant | Oct 2026
- **Project Management: Energy and Water Projects** | PMI / LinkedIn Learning | Feb 2026
- **Occupational Safety and Health: Hazard Communication** | LinkedIn Learning | Feb 2026
- **Approved Engine Course (AEC 1)** | Maritime Professional Training | Feb 2024
- **STCW Basic Safety Training** | Professional Yachtmaster Training | Dec 2023
- **LEED Green Associate Exam Preparation** | LinkedIn Learning | Sep 2026 (exam preparation course)
- **AI for Project Management: Managing Risk with Generative AI** | NASBA / LinkedIn Learning | Feb 2026

LANGUAGES

English (native) | Afrikaans (proficient) | Hebrew (basic)

SCAN OR CLICK TO CONNECT

Contact details, portfolio & research

Opens my portfolio page with project summaries, publication links and contact details.